

Lab 2 - Linguistics Data, Stat 215A, Fall 2025

October 3, 2025

1 Introduction

Dialects reflect how language changes in response to region, community, and history, and they provide insights into both social identity and cultural transmission [1], [2]. Traditional dialectology focused on mapping specific linguistic features, such as pronunciations, word choices, or constructions, onto regions. Yet individual features rarely correspond neatly to geographic boundaries. To address this, dialectometry emerged and offers a quantitative framework that aggregates across many linguistic variables to identify patterns of similarity and variation at scale. This computational approach allows to move beyond anecdotal or individual examples toward a systematic understanding of how geography and language interact.

The dataset examined in this report originates from a large-scale survey of lexical and phonological variants collected across numerous U.S. cities. Each respondent answered dozens of linguistic questions (for example, preferred words for common concepts), producing a structured dataset with both geographic coordinates and categorical responses. The goal of this analysis is to explore whether dialectometric methods can uncover regional linguistic clusters, and to evaluate the robustness of these clusters using modern diverse clustering techniques.

This report investigates three main questions:

1. What broad geographic patterns of linguistic similarity exist in the dataset?
2. Can dimensionality reduction methods reveal low-dimensional structures that highlight these regional differences?
3. How stable are the detected clusters under perturbations of the data?

This report is structured as follows. Chapter 2 describes the linguistic dataset, its variables, and the preprocessing choices, with particular attention to cleaning and preparation. Chapter 3 applies dimension reduction methods such as PCA, NMF, and t-SNE to visualize linguistic variation in a reduced space, highlighting the major axes of differentiation. Chapter 4 examines clustering results using k-means, hierarchical clustering, and related techniques to identify potential dialect regions. Chapter 5 assesses the robustness of these findings by perturbing the dataset and re-running analyses, testing how sensitive the results are to small changes. Chapter 6 concludes by situating the findings in the broader context of data science, dialectometry, and linguistic variation, and by proposing reality checks for validating clustering results.

2 The Data

The dataset used in this analysis comes from the online survey “Dialect Survey: Data and Methods” conducted by Bert Vaux and colleagues, which is available at <https://www.dialectsofenglish.com/> and with the analysis and questions at <http://dialect.redlog.net>. The survey collected responses from speakers across the United States to a wide range of lexical and phonological questions (for example, what word is used for “carbonated soft drink”). In this analysis we focus on the questions that address lexical differences, numbered 50 - 121.

The original dataset, `lingData.txt`, contains the raw survey responses for 47,471 respondents across the United States. Each row corresponds to one respondent and includes the variables `ID`, `CITY`, `STATE`, `ZIP`, `Q050-Q121` (with some questions in this range omitted), as well as `lat` and `long`. `ID` identifies the respondent, while `CITY` and `STATE` were self-reported. Latitude and longitude were later added by mapping the reported `ZIP` codes to their central coordinates. The variables beginning with `Q` represent answers to the corresponding survey questions. A value of 0 indicates no response, while positive integers correspond directly to the answer choices listed on the survey website (e.g., a value of 1 indicates answer choice (a)).

Using this base-dataset, `lingLocation.txt`, is an aggregated version of the survey responses. Here, the cate-

gorical responses were converted to binary indicators and then binned into one-degree latitude by one-degree longitude squares. Within each bin, the binary response vectors were summed across all individuals in that geographic square. For example, if two respondents in the same grid cell gave different answers, the aggregated row in `lingLocation.txt` would contain counts of each answer choice, without normalization. Thus, this dataset summarizes the distribution of responses geographically.

`question_data.RData` contains the text of the questions and the corresponding answer options. This file serves as a key for interpreting the numeric encodings in `lingData.txt` and `lingLocation.txt`.

These data are directly relevant to the problem of interest because they link linguistic variation to geography in a structured form. The combination of survey responses and location information allows us to apply statistical techniques to explore regional patterns of lexical choice, and thus to investigate the broader question of how language varies geographically across the United States.

2.1 Data Cleaning

The dataset was processed through a cleaning pipeline to repair geographic information and to handle missing survey responses. Substantial missingness was present in the geographic fields of in `ling_data`: prior to cleaning there were 540 missing `CITY`, 3 missing `STATE`, and 1,020 missing values in both `lat` and `long`.

First, rows with empty city, state, and coordinate information, were identified and imputed. Two complementary strategies were implemented:

1. When `CITY/STATE` (and typically `ZIP`) were present but coordinates were missing, latitude and longitude were imputed by matching the `ZIP` code (or city–state pair) to its central coordinates.
2. When coordinates were present but `CITY/STATE` were missing, place names were inferred by reverse-mapping from coordinates. A spatial join against `state_gdf` yielded the state, and the nearest named place was assigned as the city. Coordinates were rounded to three decimals to stabilize duplicates and align with later one-degree binning.

After reconstruction, missing entries for `CITY` and `STATE` were reduced to 0, while 154 rows with missing latitude/longitude remained. These irreparable rows were dropped using `drop_missing_coords()`, leaving 47,317 respondents in the dataset.

Subsequently, text fields were standardized with `normalize_city_state()`, ensuring consistency in capitalization (e.g., “berkeley” → “Berkeley”). Finally, survey response codes of 0, which denoted nonresponse in the original instrument, were recoded to `NaN`, so that nonresponses are treated as missing values rather than as valid categories.

`lingLocation.txt` did not contain missing values and was therefore left untouched. Since this dataset was created through aggregation of cleaned individual responses into geographic bins, it was considered valid to assume that the file was already consistent and suitable for analysis without further modifications.

2.2 Exploratory Data Analysis

As a first step, the distribution of responses to a single question from `lingData` was visualized. Figure 1 shows the geographic variation in responses to Question 105, which asked respondents for their generic term for a sweetened carbonated beverage. The plot illustrates that the dataset contains different densities of responses across the United States, with some regions heavily represented while others have relatively few data points. Regional patterns and clustering are already visible: for example, “soda” dominates in the Northeast, “pop” is common in the Midwest, and “coke” is frequently chosen in the South. These clear regional differences demonstrate that the data is well suited for investigating dialect variation.

At the same time, the figure is heavily overplotted, which makes it difficult to interpret patterns at a finer scale. This arises from the fact that thousands of individual points are plotted directly, often overlapping in areas with high respondent density. To address this, an aggregated representation will be considered next. The file `lingLocation.txt` groups responses into $1^\circ \times 1^\circ$ latitude–longitude grid cells and records counts of each answer within a cell.

To reduce the overplotting seen in the categorical scatter plots of `lingData`, responses were aggregated into $1^\circ \times 1^\circ$ latitude–longitude grid cells in the `lingLocation.txt` dataset, cf Section 2. For each cell, the most frequent (“winning”) answer was determined and plotted on the map as a first step into dimensionality reduction. This method provides a clearer view of regional distributions by smoothing over individual variation while preserving dominant local preferences. A limitation of the winner–takes-it–all representation is that it

Q105: What is your generic term for a sweetened carbonated beverage?

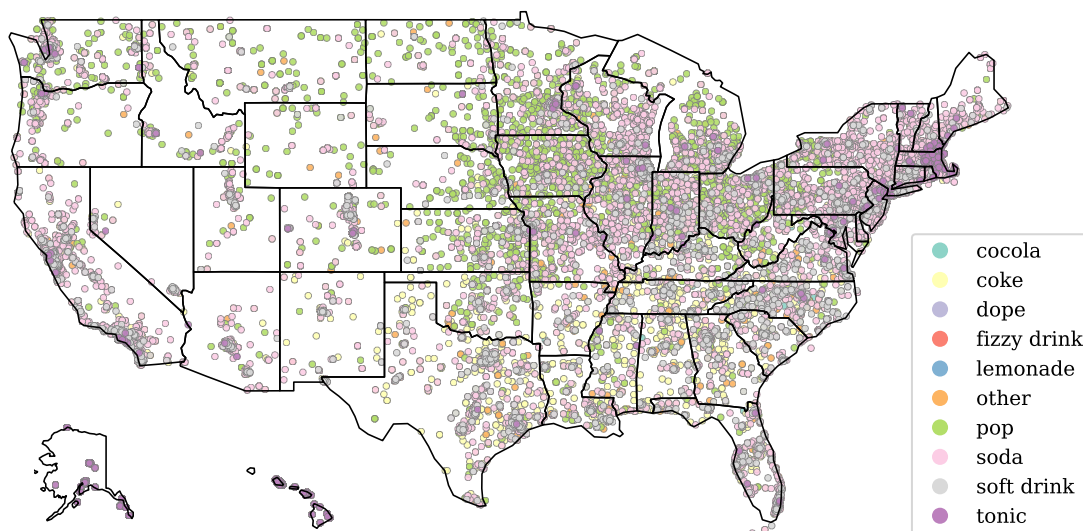
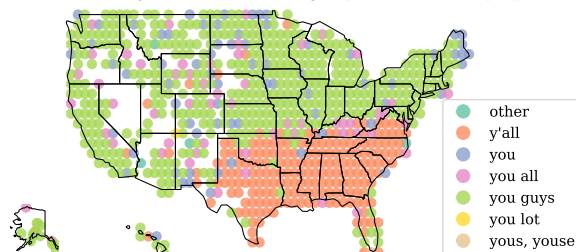


Figure 1: Geographic distribution of responses to Question 105: “What is your generic term for a sweetened carbonated beverage?” Each point represents an individual respondent from the `lingData` dataset. Clear regional patterns are visible (e.g., “soda” in the Northeast, “pop” in the Midwest, and “coke” in the South), although the plot is heavily overplotted in areas with high respondent density.

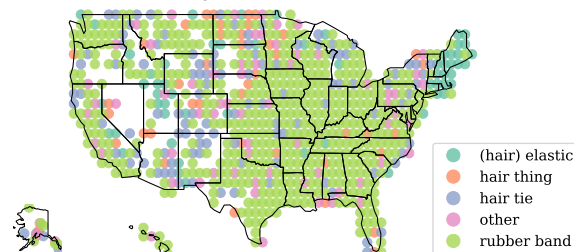
ignores within-cell variation. If two variants occur in nearly equal proportions, the cell is still colored by a single “winner,” which can exaggerate sharp boundaries or create misleading clusters.

What word(s) do you use to address a group of two or more people?



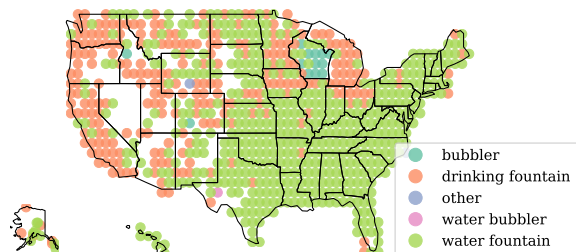
(a) Q50: How to address a group of two or more people.

What is the thing that women use to tie their hair?



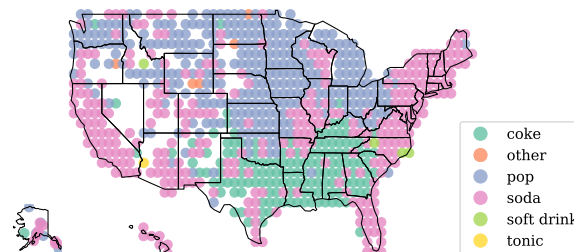
(b) Q85: Word for a hair tie.

What do you call the thing from which you might drink water in a school?



(c) Q103: Term for a drinking fountain.

What is your generic term for a sweetened carbonated beverage?



(d) Q105: Generic term for a sweetened carbonated beverage.

Figure 2: Predominant (“winner”) responses by $1^\circ \times 1^\circ$ grid cell across the U.S. for selected survey questions. (a) A sharp contrast is visible between the South and the rest of the country, with the North and West forming a broad cluster. (b) The Northwest states (Maine) show one dominant pattern, while other regions display a different, more uniform distribution. (c) A highly localized cluster appears in Wisconsin, contrasting with broader regional patterns in the Northwest and the Southeast. (d) Clear clusters appear in the Northeast, Midwest, and South, with additional concentration on the West Coast.

The maps reveal distinct regional clusters. For Question 105 (generic term for a sweetened carbonated beverage), the trend from Figure 1 become clearer and easier to see. “Soda” dominates the Northeast and parts of the West Coast, “pop” is clustered in the Midwest, and “coke” is spread across the South. Similar regional preferences appear for other questions: for Question 50 (how to address a group of people), “you guys” is predominant in much of the North and West, while “y’all” is strongly clustered in the South. Question 85 (term for a hair tie) shows “rubber band,” to be common in the South and Midwest, and a very specific cluster of “(hair) elastic” in the Northeast Maine area. Question 103 (term for a drinking fountain) highlights the unique use of “bubbler” in Wisconsin and contrasting “drinking fountain” and “water fountain” between Northwest and Southeast.

Responses to different questions are not independent; for example, regions where “y’all” is the dominant choice for Q50 often also show “coke” as the winner for Q105, indicating that answers to one question can help predict answers to another. However, there also unexpected clusters, such as the “bubbler” state Wisconsin or ‘Maine’, where people use the word “elatic” for hair tie.

3 Dimension Reduction

Dimension reduction techniques are useful when working with high-dimensional survey data because they allow broad variation patterns to be summarized along a few components. Instead of inspecting dozens of lexical questions separately, methods like PCA compress the information into a small number of axes that capture the main sources of regional differentiation. This makes it possible to visualize and interpret overall dialect structure, while also reducing noise. To build intuition, I first apply PCA to single questions that show well-known regional variation, and then extend the method to the full dataset to extract global dimensions of lexical geography.

3.1 Principal Component Analysis

Principal Component Analysis (PCA) was first applied to individual questions to illustrate how the method captures known clusters as continuous gradients. PCA reduces high-dimensional data into a small number of components that explain the largest variance. Each component is a weighted combination of the original variables, which makes it possible to summarize broad patterns of variation in just one or two axes. PCA is appropriate here because responses can be represented as continuous proportions.

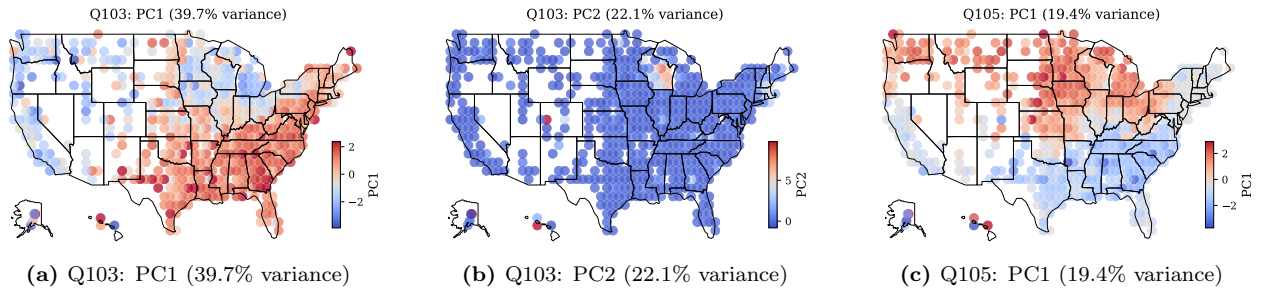


Figure 3: Principal Component Analysis for selected questions. Q103 highlights a Mid-/Northwest vs. Southeast divide (PC1) and isolates the Wisconsin cluster (PC2), while Q105 emphasizes the South vs. North/Midwest contrast, consistent with the winner maps.

For example, for Question 103 (“bubbler”), PC1 explains 39.7% of the variance and separates the Mid-/Northwest from the Southeast, whereas PC2 (22% variance) isolates the Wisconsin cluster that was already visible in the winner map (Figure 2c). The corresponding PCA plots are shown in Figure ?? . Similar results are seen for other iconic questions: Q85 (hair tie) captures a divide between South/Midwest and the rest of the country, and Q105 (carbonated beverage) highlights the contrast between the South and the North/Midwest. While these results confirm the regional clusters seen before, they also show that PCA smooths the patterns into continuous gradients rather than enforcing hard boundaries.

To capture systematic variation across all questions, PCA was then applied to the full dataset. After converting counts to row proportions, the data were standardized column-wise so that frequent and rare variants contributed equally. Cells with fewer than 20 responses were removed to avoid spurious noise. Figure 4 shows the first two components. PC1 (4.0% variance) separates the South from the North/Midwest, while PC2 (2.8% variance) highlights an East/West contrast. These broad divisions generalize the same regional patterns already observed in single-question maps, but integrate information across all lexical items.

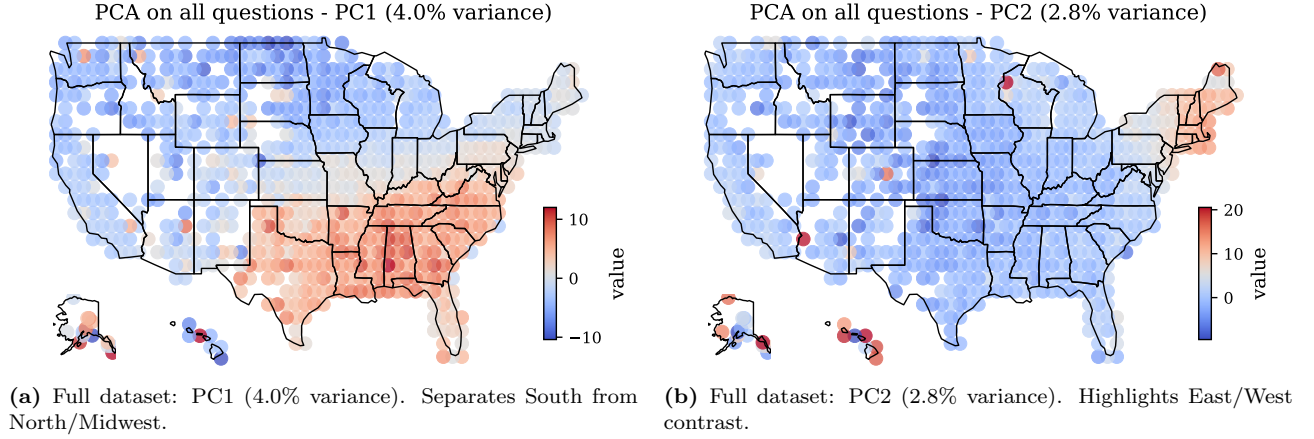


Figure 4: Principal Component Analysis on the full dataset across all lexical questions. The first two components summarize the main axes of dialectal variation in American English.

3.2 Nonnegative Matrix Factorization

In addition to PCA, Nonnegative Matrix Factorization (NMF) was applied to the data. Unlike PCA, which produces orthogonal components that capture variance contrasts, NMF decomposes the data into additive, nonnegative components. Each component can be interpreted as a “dialect profile,” emphasizing particular regional patterns rather than opposing global gradients. NMF is natural because the data are nonnegative counts.

For single questions, NMF produces components that highlight geographically coherent regions. Figure 5 shows the first NMF component for Q50, Q85, Q103, and Q105. In each case, the method emphasizes localized clusters consistent with the winner plots: the South for Q50, the Wisconsin “bubbler” region for Q103, and the South versus North/East divide for Q105. Compared to PCA, these clusters appear sharper and more interpretable, since NMF represents presence rather than contrast.

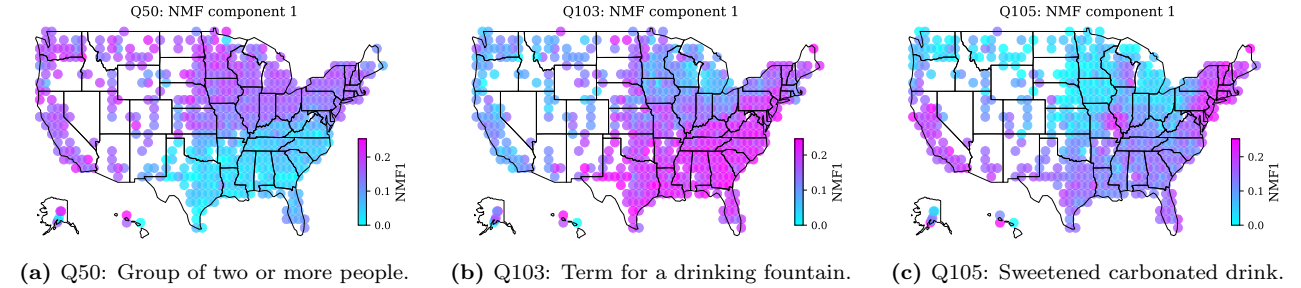


Figure 5: First NMF components for selected questions. NMF emphasizes localized clusters that align with the regional patterns from the winner plots, producing parts-based profiles rather than opposing gradients.

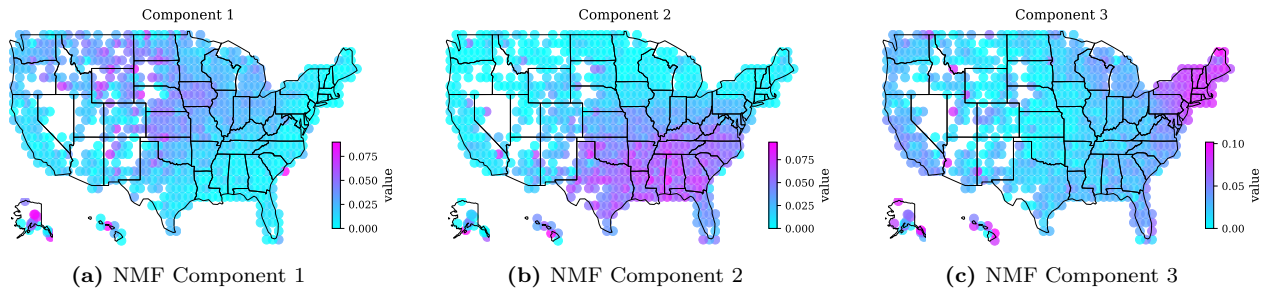


Figure 6: First three NMF components on the full dataset. Unlike the sharp clusters in the single-question case, the full-dataset components appear as smoother geographic gradients, indicating overlapping regional influences.

To capture broader patterns across all questions, NMF was also applied to the full dataset. After filtering cells with fewer than 20 responses and normalizing counts to proportions, the data were decomposed into five components. Figure 6 shows the first three components. Unlike the single-question results, these components appear less like isolated clusters and more like smooth gradients across regions. For instance, Component 1 (Figure 6a) highlights a gradual South-to-North transition, while Component 2 (Figure 6b) captures an East/West divide. Component 3 (Figure 6c) shows a weaker pattern but still emphasizes variation across the Midwest. This gradient-like behavior reflects how multiple lexical variants co-occur across regions, making NMF on the full dataset more similar in appearance to PCA, though still based on additive, nonnegative profiles.

3.3 t-distributed Stochastic Neighbor Embedding

To further explore the structure of the data, t-distributed Stochastic Neighbor Embedding (t-SNE) was applied to the response distributions. Unlike PCA, which emphasizes global variance, or NMF, which extracts additive parts-based components, t-SNE is a nonlinear technique designed to preserve local neighborhood relationships. This makes it well suited for visualizing high-dimensional categorical data in two dimensions, where clusters or continua can emerge without the constraints of linearity or orthogonality.

t-SNE was also applied to the full dataset of all lexical questions. A PCA pre-reduction to 20 dimensions was performed to denoise and speed up optimization, followed by t-SNE with perplexity set to 25. The resulting embeddings (Figure 7) show clusters that align with major geographic regions, while also highlighting finer substructure. Dimension 1 separates the South from the North/Midwest, Dimension 2 emphasizes East–West contrasts, and Dimension 3 brings out smaller regional patterns.

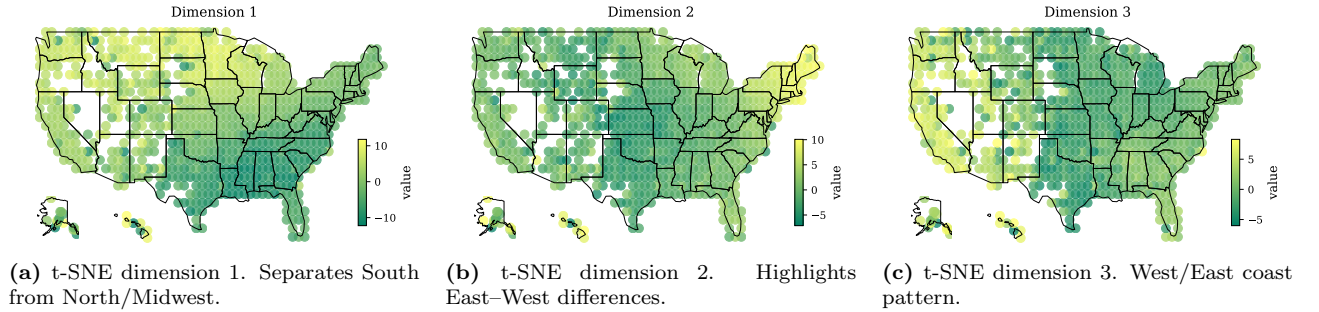


Figure 7: t-SNE on the full dataset. Compared to PCA and NMF, t-SNE emphasizes local clusters and sharpens boundaries, but at the expense of global structure.

Implementation For each analysis, response counts were normalized to row proportions. A PCA pre-reduction (20 dimensions) was applied before t-SNE for denoising and speed, and perplexity was set to 25 with PCA initialization for stability. The resulting embeddings (TSNE1, TSNE2, TSNE3) were attached back to the data and mapped over geography.

3.4 Dimension Reduction Summary

In summary, PCA highlighted broad continuous gradients of variation, NMF identified additive and interpretable regional profiles, and t-SNE emphasized fine-grained local clusters. These methods provide complementary views of the same underlying patterns of lexical differentiation. Importantly, dimension reduction transforms the high-dimensional response space into a smaller number of interpretable dimensions. This not only aids visualization but also lays the groundwork for clustering, since clusters are often more easily detected in low-dimensional spaces where major axes of variation have already been extracted. Thus, the results here provide both insight into dialect geography and a foundation for the clustering analysis that follows.

4 Clustering

While the dimension reduction methods in the previous section summarized variation along a few continuous axes, clustering aims to group locations with similar linguistic profiles into discrete sets. Instead of projecting the data onto components, clustering is performed directly on the full dataset, allowing us to identify potential dialect regions and examine whether they correspond to geographic areas. This complements the earlier analyses by shifting the focus from continuous gradients of variation to categorical groupings of speakers or locations

from the full dataset.

While dimension reduction, see Chapter 3, provides a useful low-dimensional summary of the data, the clustering analyses below are performed directly on the full normalized dataset as asked in the instructions. This ensures that the results are not dependent on a particular reduction method, although in practice dimension reduction can also be used as a preprocessing step for clustering to get more stable and less noise clustering results.

4.1 K-means clustering

K-means clustering partitions the data into k groups by minimizing the variance within clusters, assigning each cell to the nearest centroid in the feature space of variant proportions. Figure 8 shows the results for $k = 3$ and $k = 5$. In both cases, broad geographic patterns emerge that align with the regional differences observed earlier in the winner plots and PCA/NMF results.

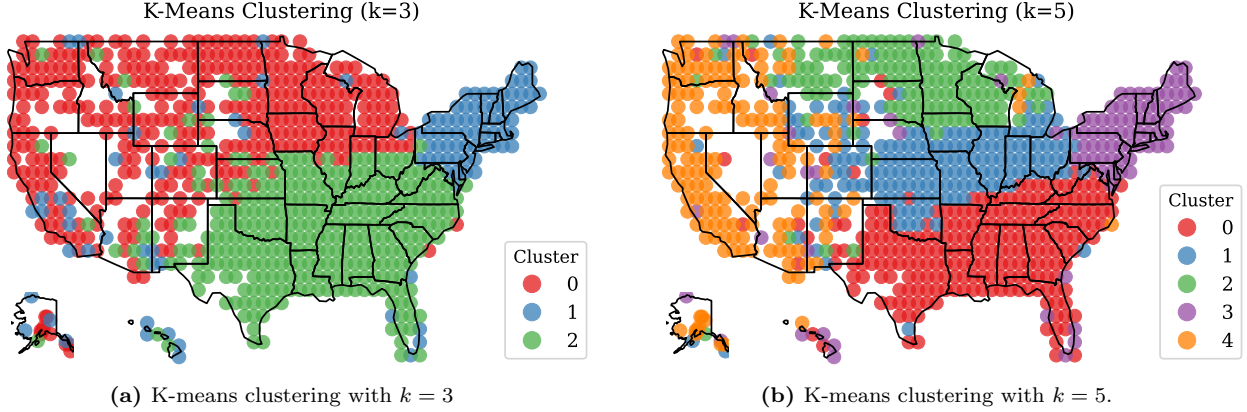


Figure 8: K-means clustering applied to the full dataset. (a) Broad clusters align with South, Midwest, and North-east/West Coast regions. (b) Subdivisions of the South and Midwest emerge, but some small scattered clusters appear. Increasing k refines clusters but also introduces smaller and less stable groups.

For example, using $k = 3$, one cluster corresponds to the South, another to the Midwest, and another to the Northeast Coast. With $k = 5$, the Midwest splits into a more specific subgroups, but some small, scattered point also appear, reflecting the sensitivity of k-means to outliers. These results suggest that while groups of respondents do align with geography, the boundaries are not completely discrete but instead form continua, particularly between the West (Cluster 4) and Clusters 0, 1, and 3. The number of clusters was chosen by experimenting with different values of k , and $k = 3$ and $k = 5$ provided the most interpretable patterns. K-means has the advantage of simplicity and interpretability, but it enforces hard, spherical clusters and can overfit isolated points, which limits its ability to capture smooth geographic transitions.

The algorithm was implemented by filtering out cells with fewer than 20 responses, normalizing counts to proportions per cell, and then running `KMeans` with multiple random initializations (`n_init=10`) for stability. The assigned cluster labels were then mapped to visualize their geographic distribution in Figure 8.

4.2 Nonnegative Matrix Factorization (NMF)

While k-means provides a simple way to partition the data into hard clusters, it assumes spherical groups in feature space and forces each location into one of these groups. An alternative approach is NMF, which represents each location not as belonging to a single cluster, but as a nonnegative combination of a small number of underlying components. In other words, instead of drawing strict boundaries, NMF identifies additive “dialect profiles” that can be mixed together to explain the observed response distributions. This is particularly appealing in the context of dialectology, since linguistic variation often reflects overlapping regional influences rather than sharply separated categories. NMF was applied to the aggregated `lingLocation` dataset to extract latent “dialect profiles.” After filtering out cells with fewer than 20 responses, the counts of each variant were row-normalized to proportions. NMF was then fitted with three and five components, using the nonnegative double singular value decomposition (`init="nndsvd"`) for stable initialization. By applying a winner-takes-all rule to the NMF components, locations can still be visualized as discrete clusters, but these clusters are grounded in the parts-based structure learned by NMF, making them more interpretable.

Figure 9 shows the resulting clusters for $k = 3$ and $k = 5$. The clusters reveal broad regional patterns similar to

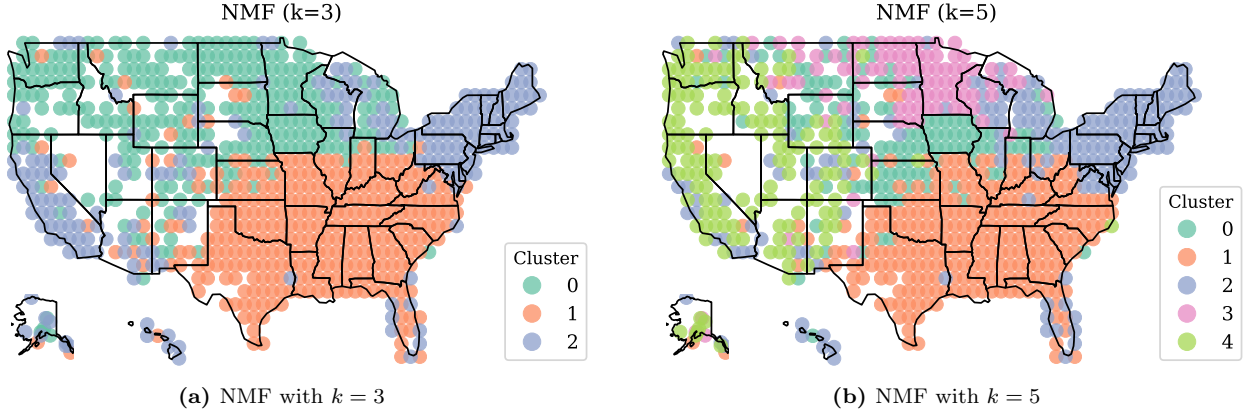


Figure 9: Winner-takes-all clustering based on NMF components. (a) . Three broad clusters emerge that align with the South, the Midwest, and the Northeast/West Coast. Compared to k-means, the boundaries are more interpretable, as they reflect additive dialect profiles rather than purely geometric partitions. (b) Increasing the number of components produces finer subdivisions, such as splitting the South into separate profiles. The clusters resemble those found with k-means but arise from parts-based components that better capture overlapping regional influences.

those observed with k-means: one group corresponds to the South, another to the Midwest, and another to the Northeast/West Coast. A key difference to k-means using $k=3$, cf. Figure 8a is that NMF identified Cluster at the west and East coast which is possible to the different mathematical approach discussed before. Increasing the number of components sharpens the clusters and introduces finer subdivisions, such as splitting the South into multiple profiles. Comparing $k=5$, i.e. Figure 8b to Figure 9b, the cluster are more similar. Compared to k-means, the NMF-based clusters are derived from a parts-based decomposition, which emphasizes additive regional profiles rather than spherical clusters in feature space. This makes the clusters more interpretable, as they can be directly linked to underlying variant distributions. However, like k-means, NMF still enforces hard assignments when using the winner-takes-all strategy, which can obscure transitional continua between regions.

4.3 Hierarchical Clustering

While NMF highlights additive dialect profiles that can be interpreted as overlapping regional influences, hierarchical clustering provides another perspective by focusing on the nested structure of linguistic similarity. Instead of fixing the number of groups from the start, hierarchical clustering builds a tree (dendrogram) that successively merges similar cells based on their response distributions. The height of the branches in the dendrogram (Figure 10) indicates the distance at which clusters were joined, and cutting the tree at a chosen level produces a desired number of clusters.

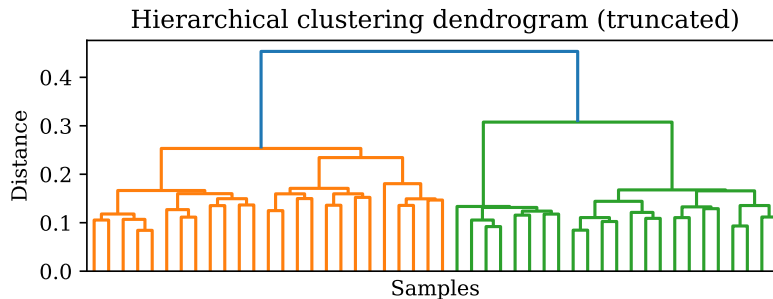


Figure 10: Dendrogram of hierarchical clustering (Ward’s method). Each merge represents a combination of similar cells, and the branch height reflects the distance at which the merge occurred. Cutting the tree at different levels yields different numbers of clusters.

In this analysis, Ward’s method was applied on the row-normalized proportion data, ensuring that merges minimize within-cluster variance. Figure 11 shows the resulting geographic maps for $k = 3$ and $k = 5$. With three clusters, broad regions emerge that align closely with the patterns already identified by k-means and NMF, namely a South cluster, a Midwest cluster, and a Northeast cluster.

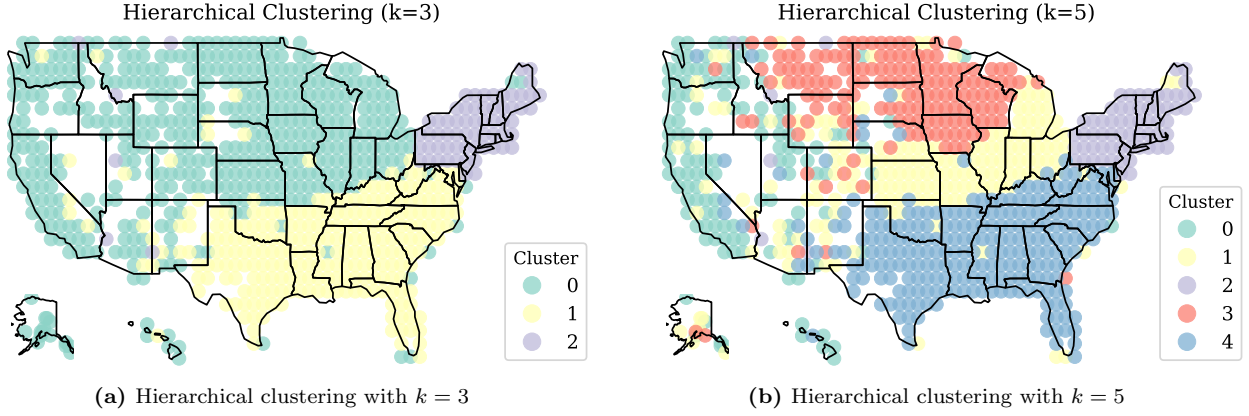


Figure 11: Hierarchical clustering of the full dataset. (a) Three broad regions emerge that correspond to South, Midwest, and Northeast, similar to earlier k-means and NMF results. (b) The South and Midwest split into more specific subgroups while the Northeast and South remain consistent.

Increasing to five clusters refines these regions further, splitting the South and Midwest into subgroups while maintaining overall coherence. Compared to NMF it emphasizes the global nesting of regions rather than additive components.

4.4 Summary of Clustering Methods

The three clustering methods applied in this analysis highlight complementary aspects of the data. K-means provided a simple baseline by partitioning the space into spherical clusters of fixed size. Its main advantage lies in its interpretability and efficiency, and in the plots with $k = 3$ and $k = 5$ it successfully recovered broad regional divisions such as the South, Midwest, and Northeast/West Coast. However, k-means was also sensitive to initialization and sometimes produced small, isolated clusters that were difficult to interpret.

NMF, in contrast, emphasized additive “dialect profiles” and offered a parts-based representation. By applying a winner-takes-all assignment, clusters similar to those of k-means emerged, but they were more directly tied to interpretable variant distributions. In the maps, NMF produced clearer and more meaningful regional clusters without the scattered outliers observed in k-means, though it still imposed hard boundaries when profiles may in reality overlap.

Hierarchical clustering added a third perspective by revealing the nested structure of regional variation. The dendrogram illustrated how clusters gradually merge, and maps with $k = 3$ and $k = 5$ produced stable regions similar to those of k-means and NMF. Compared to k-means, hierarchical clustering yielded smoother transitions and avoided isolated single-cell clusters, while compared to NMF it emphasized global structure rather than additive components.

In general all methods yielded similar clusters.

5 Stability of findings to perturbation

To assess the robustness of the k-means clustering with $k = 5$, the algorithm was re-run under two types of perturbations: changing the random seed for initialization and bootstrapping the dataset by subsampling a fraction of the grid cells. Figure 12a shows the baseline result with seed 0, which produced five clusters corresponding broadly to the South, Midwest, and Northeast/West Coast, with some finer subdivisions, cf. also Figure 8b.

When the random seed was changed (Figure 12b), the same broad clusters reappeared, although some boundary regions were reassigned, which was expected due to the sensitivity of k-means to initialization. Bootstrapping the dataset (Figures 12c and 12d) also preserved the main regional divisions, but cluster boundaries became more variable and smaller regions appeared or disappeared depending on the sample. Overall, this analysis shows that the core clusters are robust to perturbations, while finer subdivisions and boundaries are less stable, indicating that the data represent both strong regional patterns and continua where cluster membership is uncertain.

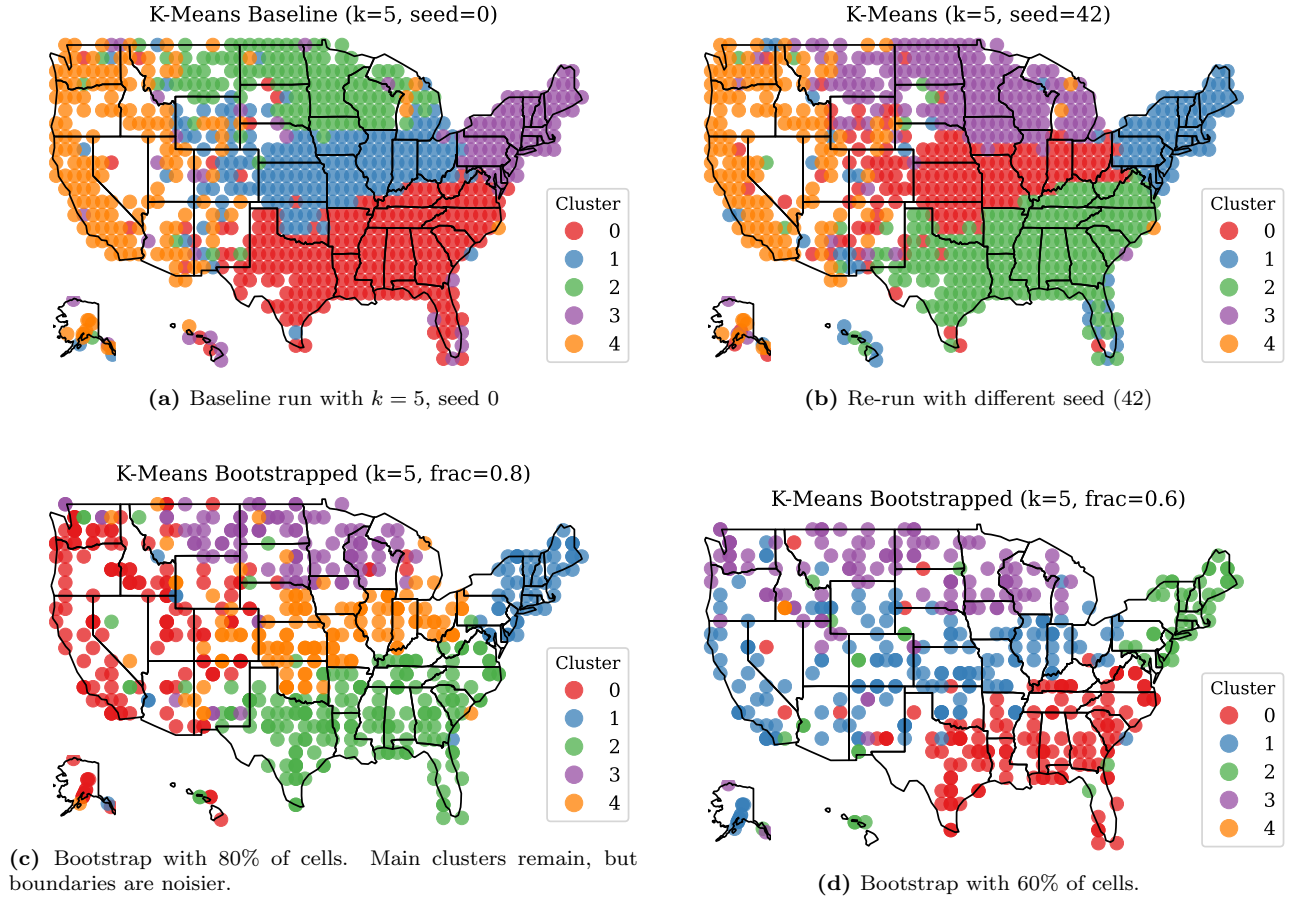


Figure 12: Stability analysis of k-means clustering with $k = 5$. The broad regional clusters are robust, while finer subdivisions and boundaries vary depending on initialization and data perturbations. Especially bootstrapping by than 60% becomes unstable.

6 Conclusion

This analysis applied dimension reduction and clustering techniques to large-scale survey data in order to investigate the geography of American English lexical variation. The three realms of data science - data, algorithms and analysis, and future data - provide a useful framework to reflect on the findings and their implications.

From the perspective of **data**, the survey provides a rich but imperfect source of information. With nearly 50,000 respondents across the United States, the dataset captures broad geographic patterns that align well with known dialect regions. At the same time, the data contain limitations: uneven sampling density across regions, self-reported location information, and missing responses that required cleaning and imputation. Aggregation into one-degree grid cells helped address overplotting and stabilized local variation, but it also introduced simplifications that may obscure fine-grained patterns. Despite these limitations, the data were sufficient to reveal robust clusters such as the South, the Midwest, and the Northeast/West Coast, and to highlight localized phenomena such as the “bubler” usage in Wisconsin.

Turning to **algorithms and analysis**, different methods offered interesting insights. Dimension reduction methods demonstrated how lexical variation could be represented as continuous structures: PCA emphasized broad gradients, NMF highlighted interpretable regional profiles, and t-SNE sharpened local clusters. Clustering methods, in turn, partitioned the data into discrete groups. K-means recovered the same broad dialect regions but was sensitive to initialization and sometimes produced spurious small clusters. NMF clustering, based on additive components, produced more “non-spherical” clusters, though the winner-takes-all assignment still enforced hard boundaries. Hierarchical clustering revealed the nested structure of linguistic similarity, showing that dialect regions can be seen at multiple levels of granularity. Stability checks confirmed that the major

regions are robust to perturbations and different random seeds, but boundaries and finer subdivisions were less stable, suggesting that dialect transitions are gradual rather than sharply separated. Taken together, these results show that the choice of method shapes how dialect boundaries are represented - either as hard partitions, overlapping profiles, or continua.

In terms of **future data**, the usefulness of this survey for decision-making depends on the context. For linguistic research, the results demonstrate the value of large-scale surveys combined with modern statistical methods, as they replicate known dialect patterns and highlight unexpected clusters. However, the uneven coverage of respondents and the age of the data mean that the findings cannot be directly generalized to the present-day U.S. population. A useful reality check would be to compare the clusters derived here with more recent large-scale data sources, such as Social Media, online surveys, or recordings from speech corpora, to verify whether the same regional patterns persist. Additionally, comparing clusters derived from lexical questions to those derived from phonetic questions could test whether the methods capture consistent dialect divisions across linguistic domains. Age dependent dataset may also provide interesting insight into changes over generations.

The main takeaways from this exploration are threefold. First, the survey data contain clear geographic signals of lexical variation that can be consistently recovered across multiple methods. Second, the choice of algorithm influences how dialect differences are represented, with k-means emphasizing discrete partitions, NMF producing additive profiles, and hierarchical clustering highlighting continua. Third, robustness analyses indicate that the strongest clusters, such as the South and the Midwest, are stable, while boundaries and finer subdivisions are more fragile and should be interpreted with caution. Given more time, further work could include experimenting with alternative clustering methods (e.g., Gaussian mixture models or density-based clustering), incorporating spatial smoothing to better capture geographic continuity, and validating the results against independent data sources. Overall, the combination of data, algorithms, and critical evaluation shows both the promise and the limitations of computational dialectology, and illustrates how statistical methods can help to understand the geography of language.

7 Bibliography

References

- [1] J. Nerbonne and W. Kretzschmar, “Progress in Dialectometry: Toward Explanation,” en, *Literary and Linguistic Computing*, vol. 21, no. 4, pp. 387–397, Sep. 2006, issn: 0268-1145, 1477-4615. DOI: 10.1093/11c/fq1034. Accessed: Sep. 24, 2025. [Online]. Available: <https://academic.oup.com/dsh/article-lookup/doi/10.1093/11c/fq1034>.
- [2] J. Nerbonne and W. Kretzschmar, “Introducing Computational Techniques in Dialectometry,” en,

8 Academic Honesty

8.1 Statement

I affirm that this report represents my own work. All sources of information, data, figures, and code that are not my own have been properly cited and acknowledged. No unauthorized assistance was received in the completion of this assignment, and I have abided by the principles of academic integrity as required by the course and the institution.

8.2 LLM Usage

An LLM (Large Language Model) was used to assist with this lab. In particular, it was helpful for writing and refining Python code for data cleaning, visualization, and clustering, as well as for improving the structure and clarity of the written report. All analyses and interpretations were ultimately carried out and verified by me, but the LLM provided useful support in streamlining coding tasks and polishing explanations.